C INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – APPLICATION BASED QUESTIONS

Course Code:	**DSA0501**
Course Title:	**Query Processing**
CO Assessed:	CO1
Assessment:	Tool 2 – Application Based Questions
Weightage:	___________
Total Marks:	20 Marks

CO1 Statement: ________________________________

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Section/Batch: Slot A 2026 Date of Submission: 27-07-2026

Faculty In-charge: K.Veena Devi

Submission Mode: Individual

Code of Conduct

I P.Sunayana 192424303_certify that this submission is my original work and have adhered to all academic integrity guidelines.

Signature: P.Sunayana

Criteria	Marks
Understanding of the Problem	4
Application of Concepts	4
Program Design / Algorithm	4
Correctness of Solution & Output	4
Explanation / Interpretation of Results	4
Total	20

O1 – AT2 – Application Based Questions

Sample Questions

Application Based Questions

Student Admission Data Cleaning (10 Marks)

A university has received a CSV file containing student admission records collected from multiple departments. The file contains duplicate records, missing phone numbers, inconsistent capitalization of department names, and extra spaces in student names. Write a Python program to read the CSV file, perform appropriate data wrangling operations to clean and standardize the data, remove duplicate records, and save the cleaned dataset into a new CSV file suitable for further analysis.

Aim

To read a student admission CSV file, clean the data by removing duplicates, filling missing phone numbers, standardizing department names, removing extra spaces from student names, and save the cleaned dataset into a new CSV file.

Algorithm

1. Start.
2. Import the pandas library.
3. Read the student admission CSV file.
4. Remove duplicate records.
5. Fill missing phone numbers with "Not Available".
6. Remove extra spaces from student names.
7. Standardize department names using proper capitalization.
8. Save the cleaned data into a new CSV file.
9. Display a success message.
10. Stop.

Code:

```
import pandas as pd
data = {
```

```
    "Student Name": [" Ravi ", "Anita", "RAHUL", " Ravi ", "Meena"],
    "Department": ["cse", "ECE", " cse ", "cse", "it"],
    "Phone": ["9876543210", None, "9123456789", "9876543210", None]
}
df = pd.DataFrame(data)
df = df.drop_duplicates()
df["Phone"] = df["Phone"].fillna("Not Available")
df["Student Name"] = df["Student Name"].str.strip()
df["Department"] = df["Department"].str.strip().str.upper()
print("Cleaned Data:")
print(df)
df.to_csv("student_admission_cleaned.csv", index=False)
```

Output:

```
Cleaned Data:
  Student Name Department          Phone
0         Ravi        CSE     9876543210
1        Anita        ECE  Not Available
2        RAHUL        CSE     9123456789
4        Meena         IT  Not Available
```

Employee Data Integration (10 Marks)

A company maintains employee details in a CSV file and department information in a JSON file. Both files contain a common Department ID field. Write a Python program to read the two files, combine the employee and department information, identify records with missing department details, and generate a consolidated dataset that can be stored in a relational database.

Aim

To combine employee and department data using Department ID, identify missing department details, and save the merged dataset.

Algorithm

1. Start.
2. Import pandas.
3. Create employee and department datasets.
4. Merge both datasets using Department ID.
5. Find records with missing department details.
6. Display the merged dataset and missing records.
7. Save the merged dataset to a CSV file.
8. Stop.

Code:

```
import pandas as pd
employee = pd.DataFrame({
    "Employee ID": [101, 102, 103, 104],
    "Employee Name": ["Arun", "Priya", "Karan", "Divya"],
    "Department ID": [1, 2, 3, 4]
})
department = pd.DataFrame({
    "Department ID": [1, 2, 3],
    "Department Name": ["HR", "Finance", "IT"]
})
merged = pd.merge(employee, department, on="Department ID", how="left")
print("Merged Data:")
```

```
print(merged)
missing = merged[merged["Department Name"].isna()]
print("\nEmployees with Missing Department Details:")
print(missing)
merged.to_csv("employee_department.csv", index=False)
```

Output:

```
Merged Data:
   Employee ID Employee Name  Department ID Department Name
0          101          Arun              1              HR
1          102         Priya              2         Finance
2          103         Karan              3              IT
3          104         Divya              4             NaN

Employees with Missing Department Details:
   Employee ID Employee Name  Department ID Department Name
3          104         Divya              4             NaN
```